Introduction to Quantum Technologies

Lectures

Dr. Ofer Kfir, room 231 @ EE labs bldg. phone: 03-6405905,
email: kfir@tauex.tau.ac.il
Reception hours – coordinated based on request.

Tutorials

Shaked Shemesh, email: shaked4@mail.tau.ac.il

Reception hours – coordinated based on request.

Important information

Final grade: final exam 85%, Project 15% (elective).

The course is designed for EE students (with or without physics courses), so we will remind ourselves with the essentials from Quantum Mechanics (QM) course.
Having written that **you need to remember the basics** we taught you on QM. If you feel rusty, we added a reminder here, and uploaded a recorded lesson with some essentials.

Project

The project is done in pairs.

There are a few options for the project, including simulations of quantum systems (Qiskit metal), or hand-on activation and manipulation of quantum bits within the Israel Quantum Computation Center (IQCC).

Lecture 1

Foundations of Quantum Mechanics

1. A quantum system is described by a state vector in Hilbert space, $|\psi\rangle$.
2. The physical properties are represented by operators. For example, the energy, E, position, x, and momentum, p, are represented by the operators $\hat{H}$ for energy, and $\hat{X}, \hat{P}$ for position and momentum).
3. When measuring a physical property, ω, is related to its eivenvalue equation ω_n are the solutions for $\hat{\Omega}|\psi_n\rangle = \omega_n|\psi_n\rangle$.
    a. The only output in an individual measurement is an eigenvalue of the corresponding operator, ω_n.
    b. The probability to measure ω_n given a state $|\psi\rangle$ is $P(\omega_n) = |\langle\psi_n|\psi\rangle|^2$.
    c. A measurement "collapses" the state to the corresponding eigenvector, $|\psi_n\rangle$.
4. The dynamics of a quantum system is governed by the Hamiltonian via the Schrödinger equation

$$i\hbar\partial_t|\psi\rangle = \hat{H}|\psi\rangle$$

The Hilbert space

- A quantum system is described by a state vector $|\psi\rangle$
- The vector-space of all possible state vectors of a system is called the *Hilbert space*.
- Hilber spaces are spanned by a basis set $\{|\phi_i\rangle\}$. For any state $|\psi\rangle$, a set of complex weights $\{c_i\}$ exist such that $|\psi\rangle = \sum_i c_i|\phi_i\rangle$
- The probability for finding a system in state $|\psi_i\rangle$ is $|c_i|^2$, where $\sum_i|c_i|^2 = 1$ is the normalization condition.
- The state vector can be expressed as "ket" $|\psi\rangle$, or "bra" $\langle\psi|$.
- The pair $\langle\psi_i||\psi_n\rangle = \langle\psi_i|\psi_n\rangle$ is the inner product of the vector space.
    - This is also referred to as the projection of $|\psi_n\rangle$ on $|\psi_i\rangle$
    - The inner product is also referred to as the overlap of the two states.
    - States with no overlap are orthogonal, $\langle\psi_i|\psi_n\rangle = 0 \rightarrow \psi_i \perp \psi_n$.
    - The inner product of the flipped vectors is complex conjugated $\langle\psi_n|\psi_i\rangle = (\langle\psi_i|\psi_n\rangle)^*$
- The norm of a state representing a physical system is $\langle\psi_i|\psi_i\rangle = 1$.
- Example: For a discrete basis set, $|\psi\rangle$ can be written as a column vector $|\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$ and $\langle\psi|$ is the conjugated row vector $\langle\psi| = (c_1^*, c_2^* \cdots c_N^*)$, where the inner product is standard linear algebra $\langle\psi_a|\psi_b\rangle = (a_1^*, a_2^* \cdots a_N^*)\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{pmatrix} = \sum_i a_i^* b_i$.

  In this example, the basis set is the trivial one

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \cdots, |\phi_N\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Operators

Operators act on the state and change it (or not), keeping it within the Hilbert space

- Can be represented as matrices $P_1 = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}, \hat{I} = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}, \sigma_x = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$
- Can be represented as ket-bra, $P_1 = |\phi_1\rangle\langle\phi_1|, \hat{I} = \sum_i |\phi_i\rangle\langle\phi_i|, \sigma_x = |\phi_1\rangle\langle\phi_2| + |\phi_2\rangle\langle\phi_1|$
  Remember: $P_1|\psi\rangle = |\phi_1\rangle\langle\phi_1|\psi\rangle = (\langle\phi_1|\psi\rangle)|\phi_1\rangle$
- For a general operator $\hat{O} = \begin{pmatrix}a & b\\ c & d\end{pmatrix} = \begin{pmatrix}O_{11} & O_{12}\\ O_{21} & O_{22}\end{pmatrix} = \sum_{nm} O_{nm}|\phi_n\rangle\langle\phi_m|$
- Adjoint operator $\left(\ \ ^{\dagger}\right)$ is the conjugated transpose.
  - For a ket the adjoint is a bra $(|\psi\rangle)^{\dagger} = \begin{pmatrix}c_1\\ c_2\\ \vdots\\ c_N\end{pmatrix}^{\dagger} = \begin{pmatrix}c_1^*\\ c_2^*\\ \vdots\\ c_N^*\end{pmatrix}^{T} = (c_1^*, c_2^* \cdots c_N^*) = \langle\psi|$
  - For an operator $\hat{O}^{\dagger} = \begin{pmatrix}a & b\\ c & d\end{pmatrix}^{\dagger} = \begin{pmatrix}a^* & c^*\\ b^* & d^*\end{pmatrix}$, or $\hat{O}^{\dagger} = (\sum_{nm} O_{nm}|\phi_n\rangle\langle\phi_m|)^{\dagger}$
    $= \sum_{nm} O_{nm}^*|\phi_m\rangle\langle\phi_n| = \sum_{nm}\left(\hat{O}^{\dagger}\right)_{mn}|\phi_m\rangle\langle\phi_n|$. I used the relation $O_{nm}^* = \left(\hat{O}^{\dagger}\right)_{mn}$.
- Unitary operators – preserve the norm.
  - Can represent a change between orthonormal basis sets
  - In matrix form, the columns are normalized vectors (the new basis set represented in the old coordinates or vice versa)
- Hermitian operators – real eigenvalues
  - This matters since for Hermitian operator A, the operator e^{iA} is unitary.
  - The Adjoint operator equals its adjoint $\hat{O} = \hat{O}^{\dagger}$
- Operators can be irreversible, such as the projection operator (example in 2D).
- Examples: rotation, mirror, stretching, projection, time-evolution, derivative.

Measurements

- Measurements are represented by operators.
- A result of a Measurement is one of its eigenvalues
- After a measurement there's certainty – the state is the corresponding eigenvector (aka "the collapse".
  - Represented by projection operators P_m of the eigenvalue m.
  - Remember to normalize
- Observables vs. expectation value
  - An observable is a value that can be the result of a single measurement (eigenvalue, m)
  - The expectation value of an operator $\hat{O}$ is the average, $\langle\hat{O}\rangle = \sum_m m p_m^{\hat{O}}$, where $p_m^{\hat{O}}$ is the probability to measure the observable m of for the operator $\hat{O}$.
- The expectation value:
  - Can be represented in bra-ket form as $\langle\hat{O}\rangle = \langle\psi|\hat{O}|\psi\rangle$
  - Can be considered as $\hat{O}$ acting on $|\psi\rangle$, and projected on $|\psi\rangle$
  - For the matrix representation of $\hat{O}$, say $\begin{pmatrix}O_{11} & O_{12} & \cdots\\ O_{21} & O_{22} & \\ \vdots & & \ddots\end{pmatrix}$, Can be represented as a quadratic form $\langle\psi|\hat{O}|\psi\rangle = (c_1^*, c_2^* \cdots c_N^*)\begin{pmatrix}O_{11} & O_{12} & \cdots & O_{1N}\\ O_{21} & O_{22} & & \\ \vdots & & \ddots & \\ O_{N1} & & & O_{NN}\end{pmatrix}\begin{pmatrix}c_1\\ c_2\\ \vdots\\ c_N\end{pmatrix}$

- For commuting operators A, B there is a basis set comprised of eigenvectors of both A and B.
    - Their measurement results can have simultaneous certainty
    - Commutation is $[A, B] = AB - BA$
    - Generally, the commutation relation is an operator (but sometimes it is a scalar).

The matrix element

- The representation of an operator by a matrix is done element by element $O_{ij} = \langle\psi_i|\hat{O}|\psi_j\rangle$
- It can be viewed as $\hat{O}$ acting on $|\psi_j\rangle$, and projected on $|\psi_j\rangle$, namely $\langle\psi_i|\hat{O}\psi_j\rangle$
- It can be viewed as $\hat{O}^\dagger$ acting on $|\psi_j\rangle$, and projected on $|\psi_i\rangle$, conjugated namely
$$O_{ij} = \langle\psi_i|\hat{O}|\psi_j\rangle = \left(\langle\psi_j|\hat{O}^\dagger|\psi_i\rangle\right)^* = \left(\langle\psi_j|\hat{O}^\dagger\psi_i\rangle\right)^*$$
- The term "matrix element" is sometimes used for moving a state to a particular measured vector. E.g., I act with a laser represented by $\hat{O}$, the matrix element is the expectation value that I measure a specific final state? $\langle\psi_{final}|\hat{O}|\psi_{initial}\rangle$, with a probability $\left|\langle\psi_{final}|\hat{O}|\psi_{initial}\rangle\right|^2$.

Schrödinger equation

The Hamiltonian is the operator that governs time propagation

$$\hat{H}|\psi\rangle = i\hbar\partial_t|\psi\rangle$$

Hence:

- Given $|\psi(t_0)\rangle$, the state at time $t = t_0 + \Delta t$ is $|\psi(t_0 + \Delta t)\rangle = \left(1 - \Delta t\frac{i}{\hbar}\hat{H}\right)|\psi(t_0)\rangle$
- At any other time t depends on the Hamiltonian
$$|\psi(t - t_0)\rangle = e^{-\frac{i}{\hbar}\int_{t_0}^{t}\hat{H}(t')dt'}|\psi(t_0)\rangle$$
(with some nuances on time ordering, ask me if you wish).
- The unitary operator $e^{-\frac{i}{\hbar}\int_{t_0}^{t}\hat{H}(t')dt'}$ is called the propagator $\hat{U}(t, t_0)$.

For time-independent Hamiltonians, the eigenvalue for vector is $|\psi_n\rangle$ is its energy E_n,
$$\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$$

- One can use the basis of the eigenvectors of $\hat{H}$, $|\psi_n\rangle$, to write
$$\hat{H} = E_n|\psi_n\rangle\langle\psi_n|$$
- The time propagator depends only the time difference
$$\hat{U}(t, t_0) = \hat{U}(t - t_0) = e^{-\frac{i}{\hbar}\int_{t_0}^{t}\hat{H}(t')dt'} = e^{-\frac{i}{\hbar}(t-t_0)\hat{H}}$$
- And by using the basis of $|\psi_n\rangle$, we can write
$$\hat{U}(t, t_0) = \sum_n e^{-\frac{i}{\hbar}E_n(t-t_0)}|\psi_n\rangle\langle\psi_n|$$

And the time propagation under a time-independent Hamiltonian is simple in the basis of the eigenenergy vectors

$$|\psi(t - t_0)\rangle = \sum_n e^{-\frac{iE_n(t-t_0)}{\hbar}}\langle\psi_n|\psi(t_0)\rangle|\psi_n\rangle$$

Example: basis vectors $|\phi_1\rangle = \begin{pmatrix}1\\0\end{pmatrix}, |\phi_2\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \hat{H} = E_0\begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}, |\psi(t=0)\rangle = \begin{pmatrix}\alpha\\\beta\end{pmatrix}$

$$\hat{U}(t, 0) = e^{-\frac{i}{\hbar}t\hat{H}} = \begin{pmatrix}e^{-\frac{iE_0t}{\hbar}} & 0\\0 & e^{\frac{iE_0t}{\hbar}}\end{pmatrix}$$

$$|\psi(t)\rangle = \begin{pmatrix} e^{-\frac{iE_0 t}{\hbar}} & 0 \\ 0 & e^{\frac{iE_0 t}{\hbar}} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha e^{-\frac{iE_0 t}{\hbar}} \\ \beta e^{\frac{iE_0 t}{\hbar}} \end{pmatrix}$$

$$\hat{U}(t,0)\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-\frac{iE_0 t}{\hbar}} \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ e^{\frac{iE_0 t}{\hbar}} \end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-\frac{iE_0 t}{\hbar}} \\ e^{\frac{iE_0 t}{\hbar}} \end{pmatrix}$$

The Bit (Mermin, classical bit vs quantum bit)

The quantum bit (qbit, qubit) includes two states that has a logic meaning of "1" and "0" marked here in the logic notation $|0\rangle_{logic}$ and $|1\rangle_{logic}$ respectively. Whereas a classical bit has two values, a quantum bit is a two-dimensional vector space, spanned by the two basis vectors $|0\rangle_{logic}$ and $|1\rangle_{logic}$.

These can be physical states of different spins $|0\rangle_{logic} = |\uparrow\rangle, |1\rangle_{logic} = |\downarrow\rangle$ or two energy levels $|0\rangle_{logic} = |E_1\rangle, |1\rangle_{logic} = |E_2\rangle$, or a photon in different waveguides $|0\rangle_{logic} = |1_{left}, 0_{right}\rangle, |1\rangle_{logic} = |0_{left}, 1_{right}\rangle$.

If we write the state in vector form, $|0\rangle = \binom{1}{0}, |1\rangle = \binom{0}{1}$ the qubit can be anything in the Hilbert space spanned by them, that is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \binom{\alpha}{\beta}$, with $|\alpha|^2 + |\beta|^2 = 1$.

Gates: the NOT operation, which we will denote by X, is represented by the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ such that:

$$X|0\rangle = |1\rangle, \qquad X|1\rangle = |0\rangle$$

The identity operation, I, is represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ such that:

$$I|0\rangle = |0\rangle, \qquad I|1\rangle = |1\rangle$$

These operations exist also for classical bit. For a quantum bit, qubit (or qbit), there are superposition states, for example:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |0\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Or generally, any normalized set, for example:

$$\frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}, \qquad \frac{|0\rangle + \sqrt{3}e^{i\phi}|1\rangle}{2}$$

Or, for example:

$$\alpha|0\rangle + \beta|1\rangle$$

Where the complex α and β comply with a normalized state $|\alpha|^2 + |\beta|^2 = 1$.

The qubit can be written as $|\psi\rangle$ and in the general case: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

We can apply more gates on qubits:

The phase operation Z is represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ such that:

$$Z|0\rangle = |0\rangle, \qquad Z|1\rangle = -|1\rangle$$

There is no point in stating the sign of a regular bit, for example:

$$Z|+\rangle = \frac{Z|0\rangle + Z|1\rangle}{\sqrt{2}} = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

The Hadamard operation H (not to mistake with a Hamiltonian which is also denoted by H)

$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$Y|0\rangle = i|1\rangle$

$Y|1\rangle = -i|0\rangle$

These 4 matrices span the operations' linear feild $\mathbb{R}^{2\times 2}$, so every operation on an $\mathbb{R}^2$ state is

a linear combination of these operation.

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

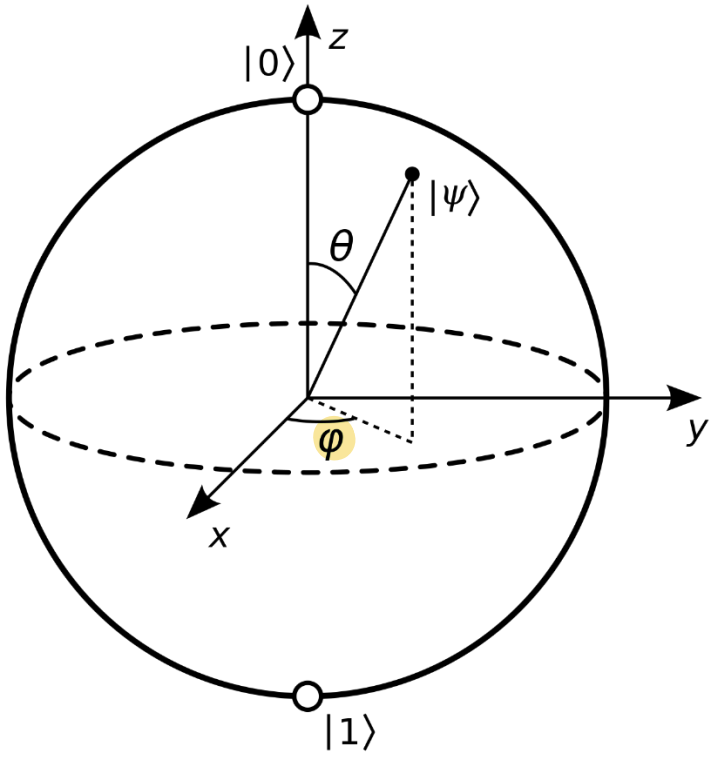
Such that:

$$H|0\rangle = |+\rangle, \qquad H|1\rangle = |-\rangle$$

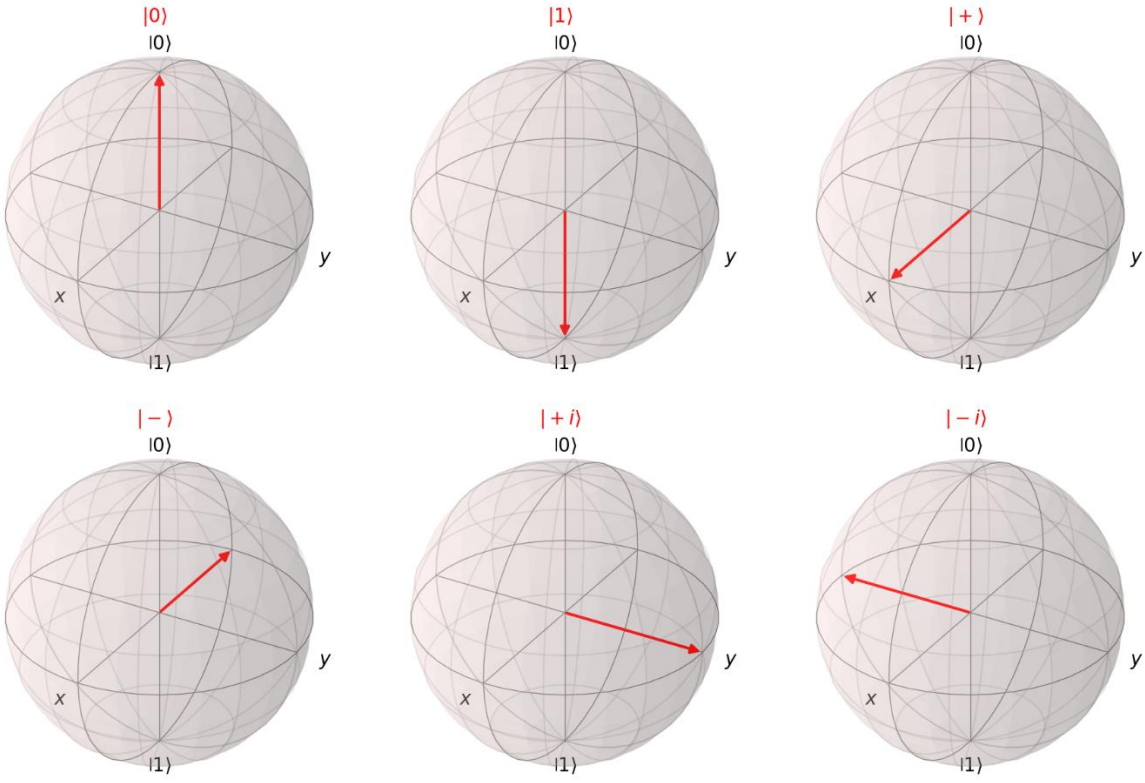
It's important to note the for all the gates above: $X^2 = Z^2 = H^2 = I$. $= \gamma^2$

The Bloch Sphere

Since the general qubit has two degrees of freedom (relative amplitudes and a relative phase, disregarding the global phase), it can be written as $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$.



- $|0\rangle$ and $|1\rangle$ are the poles on the z axis.
- $\theta = 0$ is $|0\rangle$ and $\theta = 180$ is $|1\rangle$.
- $|+\rangle$ and $|-\rangle$ are on the x axis.
- $|i\rangle$ and $|-i\rangle$ are on the y axis $\left(\theta = 90, \phi = \frac{\pi}{2}\right)$.



We will show that the Z operation is $R_z(\pi)$ and the X operation is $R_x(\pi)$, up to a $(-i)$ phase.



It's important to note that this is not exactly how the Bloch sphere changes when applying a gate.

We will find $R_y(\pi)$ and construct it from $R_x(\pi)R_z(\pi) = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$ but

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}^2 = \begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$ and we want $R_y^2(\pi) = R_x^2(\pi) = R_z^2(\pi) = (Z^2 = X^2) = I$, so we define:

$$Y \propto R_y(\pi) = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} = i \cdot R_x(\pi)R_z(\pi)$$

Note – we'll see in the next talk that the rotation is correct (i) upto a global phase, and (ii) rotations on the Bloch spheres are of half-angles.

Pauli matrices

These matrices, X, Y, Z are called typically the Pauli matrices, marked σ_m with $m = 0,1,2,3 = 0, x, y, z$

$$\sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Note

- The diagonal is dominated by I, σ_z, and the off-diagonal by σ_x, σ_y,
- $\sigma_m^2 = I$, so $\sigma_m^{2n} = I$ and $\sigma_m^{2n+1} = \sigma_i$. Useful example $e^{i\sigma_m \frac{\theta}{2}} = Icos\left(\frac{\theta}{2}\right) + \sigma_m \sin\left(\frac{\theta}{2}\right)$
- The eigenvalues of any Pauli matrix is $\lambda_\pm = \pm 1$

  Eigenvectors are,
  $$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad |\lambda_{x,\pm}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$
  $$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad |\lambda_{y,\pm}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm i \end{pmatrix}$$
  $$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad |\lambda_{z,\pm}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$
- This set comprises **any** matrix with real eigen values.

Why does it matter? Since e^{iA} is norm-preserving if A has real eigenvalues – A is Hermitian.

Time propagation

Time dynamics is given by the Schrödinger equation

$$i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$$

$$\Rightarrow |\psi(t)\rangle = e^{-\frac{i}{\hbar}\int_0^t H(t')dt'}|\psi(0)\rangle,$$

where the propagator is $U(t) = e^{-\frac{i}{\hbar}\int_0^t H(t')dt'}$.

If the Hamiltonian is time-independent, $H(t) = H$

$$\Rightarrow |\psi(t)\rangle = e^{-\frac{i}{\hbar}Ht}|\psi(0)\rangle$$

where the propagator for a time-independent Hamiltonian is $U(t) = e^{-\frac{i}{\hbar}Ht}$.

Pauli matrices as rotation generators

Example, rotation on the Bloch sphere around x $R_x(\theta) = e^{-i\sigma_x\frac{\theta}{2}}$ or if we define some axis $\hat{n} = n_x\hat{x} + n_y\hat{y} + n_z\hat{z}$, we can write any a general form for the pauli matrices as $n_x\sigma_x + n_y\sigma_y + n_z\sigma_z = \hat{n}\cdot\vec{\sigma}$.

Thus, rotation around that axis $\hat{n}$ on the bloch sphere is, $R_{\hat{n}}(\theta) = e^{-i(\hat{n}\cdot\vec{\sigma})\frac{\theta}{2}}$.

Question: show that $R_y(\theta) = e^{-i\sigma_y\frac{\theta}{2}}$ is the rotation matrix in 2D (X-Z plane). Plot it on the Bloch sphere.
Use the relation $\sigma_y^2 = I$

Answer: $e^{-i\sigma_y\frac{\theta}{2}} = I + \left(-i\sigma_y\frac{\theta}{2}\right) + \frac{1}{2}\left(-i\sigma_y\frac{\theta}{2}\right)^2 + \frac{1}{3!}\left(-i\sigma_y\frac{\theta}{2}\right)^3 \ldots$

$$= I + -i\sigma_y\frac{\theta}{2} - \frac{1}{2}\left(\frac{\theta}{2}\right)^2 - i\sigma_y\left(-\frac{1}{3!}\left(\frac{\theta}{2}\right)^3\right) \ldots = I\cos\left(\frac{\theta}{2}\right) - i\sigma_y\sin\left(\frac{\theta}{2}\right)$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cos\left(\frac{\theta}{2}\right) + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \sin\left(\frac{\theta}{2}\right) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$R_y(\theta)$ is the real-valued rotation matrix.

Question: what angles θ of $R_y(\theta)$ should be used to transform $|0\rangle$ to $|+\rangle$ and $|-\rangle$. Plot it on the Bloch sphere.

Answer: the vector form of $|0\rangle$ $|+\rangle$ and $|-\rangle$ is $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $|-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

For $\theta = \frac{\pi}{2}$ one get $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and for $\theta = -\frac{\pi}{2}$, $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Question: represent explicitly the rotation matrix $R_x(\theta)$, and find the states it forms when rotating $|0\rangle$ by $\pm\frac{\pi}{2}$ and $\pm\pi$

Answer: Similar to the above, $R_x(\theta) = e^{-i\sigma_x\frac{\theta}{2}} = I\cos\left(\frac{\theta}{2}\right) - i\sigma_x \sin\left(\frac{\theta}{2}\right)$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cos\left(\frac{\theta}{2}\right) + \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \sin\left(\frac{\theta}{2}\right) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$R_x(\theta)$ is the symmetric rotation matrix.

$$R_x\left(\pm\frac{\pi}{2}\right)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \mp i \end{pmatrix} = |\mp i\rangle\,, \qquad R_x(\pm\pi)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \mp i\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \mp i|1\rangle$$

So $R_x\left(\pm\frac{\pi}{2}\right)|0\rangle = |\mp i\rangle$ (different states) and $R_x(\pm\pi)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \mp i|1\rangle$ (same state, π-phase).

Example: representing a diagonal Hamiltonian H_0 as a rotation.

For a Hamiltonian in a given basis, where

$$\langle\psi_1|\hat{H}|\psi_1\rangle = E_a\,, \qquad \langle\psi_2|\hat{H}|\psi_2\rangle = E_b, \qquad \langle\psi_1|\hat{H}|\psi_2\rangle = \langle\psi_2|\hat{H}|\psi_1\rangle = 0,$$

and $|\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\psi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

If we mark $\bar{E} = \frac{1}{2}(E_a + E_b)$, $\Delta E = \frac{1}{2}(E_a - E_b)$

$$H = \begin{pmatrix} E_a & 0 \\ 0 & E_b \end{pmatrix} = \bar{E}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \Delta E\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \bar{E}I + \Delta E\sigma_z$$

$$U(t) = e^{-\frac{i}{\hbar}\hat{H}t} = e^{-\frac{i}{\hbar}\bar{E}t}e^{-i\frac{\Delta E t}{2\hbar}\sigma_z} = e^{-\frac{i}{\hbar}\bar{E}t}R_z\left(\frac{\Delta E}{\hbar}t\right)$$

→ time propagation is z-rotation !

(*) Numerical examples for rotations on the Bloch sphere: $(\sigma_x, \sigma_y, \sigma_z, \sigma_z + \sigma_y)$ and H_0 rotations.

links:

https://www.st-andrews.ac.uk/physics/quvis/simulations_html5/sims/blochsphere/blochsphere.html

https://www.bloch-sphere.app/

https://www.falstad.com/qm1drad/

Generally, a two-level Hamiltonian can be represented as, $\hat{H} \propto \vec{n} \cdot \vec{\sigma}$.

$$\langle\psi_1|\hat{H}|\psi_1\rangle = E_a\,, \qquad \langle\psi_2|\hat{H}|\psi_2\rangle = E_a,$$

$$\langle\psi_1|\hat{H}|\psi_2\rangle = r^*, \qquad \langle\psi_2|\hat{H}|\psi_1\rangle = r$$

$$H = \begin{pmatrix} E_a & r^* \\ r & E_b \end{pmatrix} = \bar{E}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \Delta E\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \Re e[r]\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \Im m[r]\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$= \bar{E}I + \Delta E\sigma_z + \Re e[r]\sigma_x + \Im m[r]\sigma_y$$

So for $\vec{n} = (\Re e[r], \Im m[r], \Delta E)$ we get $\quad H = \bar{E}I + \vec{n} \cdot \vec{\sigma}$.

Note that the mean energy is arbitrary, and upon a choice of $\bar{E} = 0$, we get that for a qubit

$$H = \vec{n} \cdot \vec{\sigma}$$

Question: what will the dynamics be for state $|\psi(t)\rangle$ under such Hamiltonian?
Starting from $|0\rangle$:
a) what is the state maximally different state it can get, and what time it is received?
b) what is the condition to allow $\langle 1|\psi(t)\rangle = 1$?

Answer: the state will rotate around axis $\vec{n}$, $R_{\hat{n}}(\theta) = e^{-i(\hat{n}\cdot\vec{\sigma})\frac{\theta}{2}}$.

Using Schrödinger $i\hbar\partial_t|\psi\rangle = H|\psi\rangle = (\vec{n} \cdot \vec{\sigma})|\psi\rangle$,

so $$|\psi(t)\rangle = e^{-\frac{i}{\hbar}Ht}|\psi_0\rangle = e^{-\frac{i}{\hbar}(\vec{n}\cdot\vec{\sigma})t}|\psi_0\rangle = e^{-i(\hat{n}\cdot\vec{\sigma})\frac{\theta}{2}}|\psi_0\rangle$$

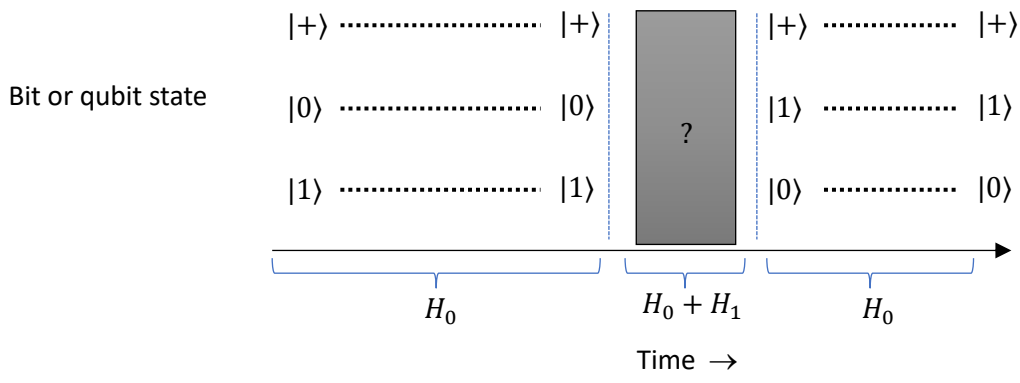
With $\theta = 2\frac{|\vec{n}|}{\hbar}t$.

a) Since $e^{-i(\hat{n}\cdot\vec{\sigma})\frac{\pi}{2}} = Icos\left(\frac{\pi}{2}\right) - i(\hat{n}\cdot\vec{\sigma})\sin\left(\frac{\pi}{2}\right) = -i\begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{pmatrix}$,
the furthest state will be for $\theta = \pi$, for $t_{max} = \frac{\pi}{2}\left(\frac{|\vec{n}|}{\hbar}\right)^{-1}$, $|\psi(t_{max})\rangle = -i\begin{pmatrix} n_z \\ n_x + in_y \end{pmatrix}$.
b) Hence, **only** rotation that lacks n_z (no σ_z in H) allows to bring $|0\rangle \overset{time}{\Longrightarrow} |1\rangle$.

<u>The time-evolution frames:</u>
In this part, we do as follows.

1) Split the Hamiltonian to the undisturbed Hamiltonian $\hat{H}_0$ and a disturbance $\hat{H}_1$.
2) Explain the importance of ignoring the dynamics of $\hat{H}_0$.
3) Show the dynamics according to an $\hat{H}_1(t)$ that depends on a frequency ω (Heisenberg picture)
4) Find a nice-looking matrix for the dynamics (rotating frame)

First point: We term $\hat{H}_0$ and a disturbance $\hat{H}_1$, such that the bit essentially unchanged under $\hat{H}_0$, and $\hat{H}_1$ is a short-lived operation:



For a qubit:

$$H = \begin{pmatrix} E_a & r^* \\ r & E_b \end{pmatrix} = H_0 + H_1, \qquad H_0 = \begin{pmatrix} E_a & 0 \\ 0 & E_b \end{pmatrix}, \qquad H_1 = \begin{pmatrix} 0 & r^* \\ r & 0 \end{pmatrix}$$

Or

$$H = \bar{E}I + \vec{n} \cdot \vec{\sigma}\,, \qquad H_0 = \bar{E}I + \Delta E \sigma_z, \qquad H_1 = \Re e[r]\sigma_x + \Im m[r]\sigma_y$$

The Typical form for a H_0 Hamiltonian is

$$H_0 = -\frac{\hbar\omega}{2}\sigma_z$$

Second, what happens under H_0

According to the Schrödinger equation $i\hbar\partial_t|\psi\rangle = \widehat{H}_0|\psi\rangle$, so the bit states are acquiring phase. If $|0\rangle$ and $|1\rangle$ are the eigenstates of H_0, such that

$$|0\rangle = |\psi_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad H_0|\psi_0\rangle = -\frac{\hbar\omega}{2}|\psi_0\rangle$$

$$|1\rangle = |\psi_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad H_0|\psi_1\rangle = \frac{\hbar\omega}{2}|\psi_1\rangle$$

Then, time propagation applies each Eigenstate with a phase $e^{-i\frac{E}{\hbar}t}$

$$|0\rangle \overset{time\ t}{\Longrightarrow} e^{i\frac{\omega t}{2}}|0\rangle \ , \qquad |1\rangle \overset{time\ t}{\Longrightarrow} e^{-i\frac{\omega t}{2}}|1\rangle \ ,$$

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \overset{time\ t}{\Longrightarrow} \frac{e^{i\frac{\omega t}{2}}|0\rangle + e^{-i\frac{\omega t}{2}}|1\rangle}{\sqrt{2}} = e^{i\frac{\omega t}{2}}\left(\frac{|0\rangle + e^{-i\omega t}|1\rangle}{\sqrt{2}}\right)$$

That is, the bits are never fixed in this Schrödinger picture of $|\psi\rangle$!

We would like to have a situation where our logical bits are fixed, something like $i\hbar\partial_t|\psi\rangle_{logic} = 0$, i.e., $|0\rangle_{logic} \overset{time\ t}{\Longrightarrow} |0\rangle_{logic}$, $|1\rangle_{logic} \overset{time\ t}{\Longrightarrow} |1\rangle_{logic}$.

Third: we would see the convenience of distinguishing between three approaches for investigating the dynamics:

- Schrodinger picture (standard, $|\psi\rangle$ under $\widehat{H}_0$, $i\hbar\partial_t|\psi\rangle = \widehat{H}_0|\psi\rangle$)

- operators are static

$$\hat{O}_{Schrödinger}$$

- Heisenberg picture ($|\tilde{\psi}\rangle$ rotates with $\hat{H}_0$, so, $i\hbar\partial_t|\tilde{\psi}\rangle = 0$)
  - Operators are time dependent

$$\hat{O}_{Heisenberg} = e^{\frac{i}{\hbar}H_0 t}\hat{O}_{Schrödinger}e^{-\frac{i}{\hbar}H_0 t}$$

- Rotating frame ($|\bar{\psi}\rangle$, rotates with a driving field at frequency ω
  - Operators are time dependent

$$\hat{O}_{rotating\ frame} = e^{i\omega t\cdot\sigma_z}\hat{O}_{Schrödinger}e^{-i\omega t\cdot\sigma_z}$$

Next, we see explicitly the dynamics in these frames

The dynamics in the Heisenberg picture and $\widehat{H_1}_{Heisenberg}$

We define the state in the Heisenberg picture as

$$|\tilde{\psi}\rangle = e^{\frac{i}{\hbar}H_0 t}|\psi\rangle$$

Let us see that under H_0, this is a static state , $\partial_t|\tilde{\psi}\rangle = 0$ (i.e., $H_1 = 0$).

$$\partial_t|\tilde{\psi}\rangle = \partial_t\left(e^{\frac{i}{\hbar}H_0 t}|\psi\rangle\right) = \frac{i}{\hbar}H_0 e^{\frac{i}{\hbar}H_0 t}|\psi\rangle + e^{\frac{i}{\hbar}H_0 t}\partial_t|\psi\rangle = \frac{i}{\hbar}H_0\underbrace{e^{\frac{i}{\hbar}H_0 t}|\psi\rangle}_{|\tilde{\psi}\rangle} + e^{\frac{i}{\hbar}H_0 t}\frac{1}{i\hbar}H_0|\psi\rangle$$

$$= \left(\frac{i}{\hbar}H_0 + \frac{1}{i\hbar}H_0\right)|\tilde{\psi}\rangle = 0$$

Intuitively, you can see that the term $e^{\frac{i}{\hbar}H_0 t}$ will counteract the effect of the time-propagation of H_0 alone, leaving the state unchanged. Let's see that with a disturbance, using $H = H_0 + H_1$

$$\partial_t|\tilde{\psi}\rangle = \partial_t\left(e^{\frac{i}{\hbar}H_0 t}|\psi\rangle\right) = \frac{i}{\hbar}H_0 e^{\frac{i}{\hbar}H_0 t}|\psi\rangle + e^{\frac{i}{\hbar}H_0 t}\partial_t|\psi\rangle$$

$$= \frac{i}{\hbar}H_0 e^{\frac{i}{\hbar}H_0 t}|\psi\rangle + e^{\frac{i}{\hbar}H_0 t}\frac{1}{i\hbar}\left(H_0|\psi\rangle + \underbrace{H_1|\psi\rangle}_{nonzero}\right)$$

And substituting $e^{-\frac{i}{\hbar}H_0 t}|\tilde{\psi}\rangle = |\psi\rangle$, we get

$$\partial_t|\tilde{\psi}\rangle = \frac{1}{i\hbar}e^{\frac{i}{\hbar}H_0 t}H_1 e^{-\frac{i}{\hbar}H_0 t}|\tilde{\psi}\rangle$$

Hence, for the Heisenberg picture:

- If $H_1 = 0$, $i\hbar\partial_t|\tilde{\psi}\rangle = 0$ – the state of a bit is not rotating without perturbation
- If $H_1 \neq 0$, to understand it's action, one needs to calculate $\frac{1}{i\hbar}e^{\frac{i}{\hbar}H_0 t}H_1 e^{-\frac{i}{\hbar}H_0 t}$.
- In the Heisenberg picture, the <u>state is fixed and the operators are time-varying</u>: any operator is $\hat{O}_{Heisenberg} = e^{\frac{i}{\hbar}H_0 t}\hat{O}_{Schrödinger}e^{-\frac{i}{\hbar}H_0 t}$

For example, any matrix element is

$$\langle\psi_i|\hat{O}_{Sch}|\psi_j\rangle = \left\langle \psi_i \middle| e^{-\frac{i}{\hbar}H_0 t} \underbrace{\left(e^{\frac{i}{\hbar}H_0 t}\hat{O}_{Sch}e^{-\frac{i}{\hbar}H_0 t}\right)}_{\hat{O}_{Heisenberg}} e^{\frac{i}{\hbar}H_0 t} \middle| \psi_j \right\rangle = \langle\tilde{\psi}_i|\hat{O}_{Heis}|\tilde{\psi}_j\rangle$$

- For a qubit system, with a Hamiltonian $H_0 = -\frac{\hbar\nu}{2}\sigma_z$, the Heisenberg picture is
$$|\tilde{\psi}\rangle = e^{-i\frac{\nu t}{2}\sigma_z}|\psi\rangle.$$
It essentially counter-rotates at the transition frequency to nullify the ν-rotation of the qubit under H_0.
- At t=0, the frames the state is represented the same way $|\tilde{\psi}\rangle_{t=0} = |\psi\rangle_{t=0}$.
- Eigenstates of H_0 differ in the Heisenberg picture only by a phase, $e^{\frac{i}{\hbar}H_0 t}|\psi_j\rangle = e^{\frac{i}{\hbar}E_j t}|\psi_j\rangle$, so the projection probability on an eigenstate is time-independent and frame-independent.

In the future, we will typically represent the state in **some** rotating frame, either Heisenberg or something else, also as $|\psi\rangle$, without the tilde. If you are unsure – ask!

The dynamics in the rotating frame and $\widehat{H_1}_{rotating\ frame}$

Consider $H_0 = -\frac{\hbar\nu}{2}\sigma_z$, and some H_1, but for a reason shown later, we choose a frame at a different frequency , ω, for which

$$|\bar{\psi}\rangle = e^{-i\frac{\omega t}{2}\sigma_z}|\psi\rangle$$

$$\partial_t|\bar{\psi}\rangle = \partial_t\left(e^{-i\frac{\omega t}{2}\sigma_z}|\psi\rangle\right) = -i\frac{\omega}{2}\sigma_z\, e^{-i\frac{\omega t}{2}\sigma_z}|\psi\rangle + e^{-i\frac{\omega t}{2}\sigma_z}\partial_t|\psi\rangle$$
$$= -i\frac{\omega}{2}\sigma_z\, e^{-i\frac{\omega t}{2}\sigma_z}|\psi\rangle + e^{-i\frac{\omega t}{2}\sigma_z}\frac{1}{i\hbar}(H_0|\psi\rangle + H_1|\psi\rangle)$$

And substituting $|\psi\rangle = e^{i\frac{\omega t}{2}\sigma_z}|\bar{\psi}\rangle$, and $\left[\sigma_z, e^{\pm i\frac{\omega t}{2}\sigma_z}\right] = 0$

$$\partial_t|\bar{\psi}\rangle = -i\frac{\omega}{2}\sigma_z\,|\bar{\psi}\rangle + e^{-i\frac{\omega t}{2}\sigma_z}\frac{1}{i\hbar}\left(\left(-\frac{\hbar\nu}{2}\sigma_z\right)e^{i\frac{\omega t}{2}\sigma_z}|\bar{\psi}\rangle + H_1 e^{i\frac{\omega t}{2}\sigma_z}|\bar{\psi}\rangle\right)$$

$$= -i\frac{\omega}{2}\sigma_z\,|\bar{\psi}\rangle + \left(\left(+i\frac{\nu}{2}\sigma_z\right)|\bar{\psi}\rangle + \frac{1}{i\hbar}e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z}|\bar{\psi}\rangle\right)$$

$$\partial_t|\bar{\psi}\rangle = \left[-i\frac{\omega-\nu}{2}\sigma_z + \frac{1}{i\hbar}e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z}\right]|\bar{\psi}\rangle$$

- $\frac{1}{i\hbar}e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z}$ is the operator of interest to calculate, given a disturbance H_1.
- In once choses $\omega = \nu$, the rotation frame is the Heisenberg frame
- The detuning $\Delta = \omega - \nu$ is the frequency of disturbance with respect to the resonance.
- Otherwise, there is an additional diagonal real term.
- At t=0 a state has the same representation in any frame
$$|\psi\rangle_{t=0} = |\tilde{\psi}\rangle_{t=0} = |\bar{\psi}\rangle_{t=0}$$
- In the absence of a disturbance, the probability of being in eigenstate of H_0, $|\psi_i\rangle$ is frame-independent
$$\partial_t|\langle\psi_i|\psi\rangle|^2 = \partial_t\left|\langle\psi_i|\tilde{\psi}\rangle\right|^2 = \partial_t\left|\langle\psi_i|\bar{\psi}\rangle\right|^2 = 0$$
- At $t \neq 0$, one should be more careful.

Heisenberg picture vs. rotating frame - the disturbance of a harmonic field

Suppose a harmonic disturbance $r(t) = r_0 e^{-i\omega t}$, such that

$$H_1(t) = \left(\Re e[r(t)]\sigma_x + \Im m[r(t)]\sigma_y\right) = \begin{pmatrix} 0 & r^*(t) \\ r(t) & 0 \end{pmatrix}$$

$$H_0 = -\frac{\hbar\nu}{2}\sigma_z = \begin{pmatrix} -\frac{\hbar\nu}{2} & 0 \\ 0 & \frac{\hbar\nu}{2} \end{pmatrix}, \qquad e^{\frac{i}{\hbar}H_0 t} = \begin{pmatrix} e^{-i\frac{\nu t}{2}} & 0 \\ 0 & e^{i\frac{\nu t}{2}} \end{pmatrix}$$

The dynamics in the Heisenberg picture is $\partial_t|\tilde{\psi}\rangle = \frac{1}{i\hbar}e^{\frac{i}{\hbar}H_0 t}H_1 e^{-\frac{i}{\hbar}H_0 t}|\tilde{\psi}\rangle$, so we need to calculate

$$\frac{1}{i\hbar}e^{\frac{i}{\hbar}H_0 t}H_1 e^{-\frac{i}{\hbar}H_0 t} = \frac{1}{i\hbar}\begin{pmatrix} e^{-i\frac{\nu t}{2}} & 0 \\ 0 & e^{i\frac{\nu t}{2}} \end{pmatrix}\begin{pmatrix} 0 & r^*(t) \\ r(t) & 0 \end{pmatrix}\begin{pmatrix} e^{i\frac{\nu t}{2}} & 0 \\ 0 & e^{-i\frac{\nu t}{2}} \end{pmatrix}$$

$$= \frac{1}{i\hbar}\begin{pmatrix} 0 & r^*(t)e^{-i\nu t} \\ r(t)e^{i\nu t} & 0 \end{pmatrix}$$

If we substitute $r(t) = r_0 e^{-i\omega t}$, and $\Delta = \omega - \nu$ we get

$$\partial_t|\tilde{\psi}\rangle = -\frac{i}{\hbar}\begin{pmatrix} 0 & r_0^* e^{i\Delta t} \\ r_0 e^{-i\Delta t} & 0 \end{pmatrix}|\tilde{\psi}\rangle$$

Δ is the detuning of the disturbance field with respect to the qubit frequency.

How would that be in the rotating frame? Let us calculate the following

$$\frac{1}{i\hbar}\boldsymbol{e^{-i\frac{\omega t}{2}\sigma_z}}H_1\boldsymbol{e^{i\frac{\omega t}{2}\sigma_z}} = \frac{1}{i\hbar}\begin{pmatrix} e^{-i\frac{\omega t}{2}} & 0 \\ 0 & e^{i\frac{\omega t}{2}} \end{pmatrix}\begin{pmatrix} 0 & r^*(t) \\ r(t) & 0 \end{pmatrix}\begin{pmatrix} e^{i\frac{\omega t}{2}} & 0 \\ 0 & e^{-i\frac{\omega t}{2}} \end{pmatrix}$$

$$= \frac{1}{i\hbar}\begin{pmatrix} 0 & r^*(t)e^{-i\omega t} \\ r(t)e^{i\omega t} & 0 \end{pmatrix} = -\frac{i}{\hbar}\begin{pmatrix} 0 & r_0^* \\ r_0 & 0 \end{pmatrix}$$

So we can substitute $\partial_t|\bar{\psi}\rangle = \left[-i\frac{\omega-\nu}{2}\sigma_z + \frac{1}{i\hbar}e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z}\right]|\bar{\psi}\rangle$ and get

$$\partial_t|\bar{\psi}\rangle = i\begin{pmatrix} -\frac{\Delta}{2} & 0 \\ 0 & \frac{\Delta}{2} \end{pmatrix}|\bar{\psi}\rangle - \frac{i}{\hbar}\begin{pmatrix} 0 & r_0^* \\ r_0 & 0 \end{pmatrix}|\bar{\psi}\rangle = -i\begin{pmatrix} \frac{\Delta}{2} & \frac{r_0^*}{\hbar} \\ \frac{r_0}{\hbar} & -\frac{\Delta}{2} \end{pmatrix}|\bar{\psi}\rangle$$

The rotating frame with a harmonic field provides for is a time-independent operator, even under detuning.

Rabi-Oscillations

For the Rabi-Oscillation one can follow Quantum Optics by Scully & Zubairy, chapter 5 Atom–field interaction – semiclassical theory, page 152. We also use Mitchel Weissbluth, Atom-photon interactions, p.118 (Rabi Formula).

Let us now show what happens to a qubit state, that, for simplicity, is initially at $|\psi\rangle_{t=0} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$, and a resonant excitation (without detuning), $r = |r_0|e^{i\phi}e^{-i\nu t}$, where $\nu = \omega$ and $\Delta = 0$.

Instead of solving $|\psi(t)\rangle$ we will solve under the Heisenberg picture, for $|\tilde{\psi}(t)\rangle$

$$\partial_t|\tilde{\psi}\rangle = -i\frac{|r_0|}{\hbar}\begin{pmatrix}0 & e^{-i\phi}\\ e^{i\phi} & 0\end{pmatrix}|\tilde{\psi}\rangle.$$

This is a simple ODE. Let us ask what the probability is for measuring the state $|0\rangle$, namely $|a(t)|^2$ for $|\tilde{\psi}(t)\rangle = \begin{pmatrix}a(t)\\ b(t)\end{pmatrix}$.

Substituting $\begin{pmatrix}a(t)\\ b(t)\end{pmatrix}$

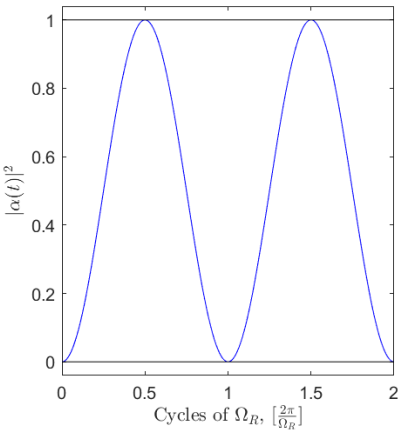
$$\partial_t\begin{pmatrix}a(t)\\ b(t)\end{pmatrix} = -i\frac{|r_0|}{\hbar}\begin{pmatrix}0 & e^{-i\phi}\\ e^{i\phi} & 0\end{pmatrix}\begin{pmatrix}a(t)\\ b(t)\end{pmatrix}$$

With $\dot{a} = -i\frac{|r_0|}{\hbar}e^{-i\phi}b$ and $\dot{b} = -i\frac{|r_0|}{\hbar}e^{i\phi}a$ one gets an equation for an oscillating amplitude

$$\ddot{a} = -\left(\frac{|r_0|}{\hbar}\right)^2 a$$

For which the $|a(t)|^2$ oscillates at a frequency $\Omega_R = 2\frac{|r_0|}{\hbar}$, which is the Rabi-frequency

$$a = \sin\left(\frac{\Omega_R t}{2}\right) \rightarrow |a(t)|^2 = \frac{1}{2} - \frac{1}{2}\cos(\Omega_R t)$$



Note we used the Heisenberg picture, but the answer is correct for the regular frame (Schrödinger) since $\left|\langle 0|\tilde{\psi}(t)\rangle\right| = |\langle 0|\psi(t)\rangle|$.

After defining Ω_R we can write the dynamics as $\partial_t|\tilde{\psi}\rangle = -\frac{i}{2}\begin{pmatrix}0 & \Omega_R e^{-i\phi}\\ \Omega_R e^{i\phi} & 0\end{pmatrix}|\tilde{\psi}\rangle$

Rabi frequency and generalized Rabi-frequency

Let us repeat the calculation for $|a(t)|^2$ for a non-zero detuning, $r = |r_0|e^{i\phi}e^{-i\omega t}$, with $\Delta = \nu - \omega$. One can solve in the Heisenberg picture the following time dependent linear equation

$$\partial_t|\tilde{\psi}\rangle = -\frac{i}{2}\begin{pmatrix}0 & \Omega_R e^{-i\phi}e^{i\Delta t}\\ \Omega_R e^{i\phi}e^{-i\Delta t} & 0\end{pmatrix}|\tilde{\psi}\rangle$$

This is easiest to solve in the rotating frame, with

$$\partial_t|\bar{\psi}\rangle = -i\begin{pmatrix}\frac{\Delta}{2} & \frac{r_0^*}{\hbar}\\ \frac{r_0}{\hbar} & -\frac{\Delta}{2}\end{pmatrix}|\bar{\psi}\rangle = -\frac{i}{2}\begin{pmatrix}\Delta & \Omega_R e^{-i\phi}\\ \Omega_R e^{+i\phi} & -\Delta\end{pmatrix}|\bar{\psi}\rangle$$

We solve for $|\bar{\psi}\rangle = \begin{pmatrix}a(t)\\ b(t)\end{pmatrix}$

$$\partial_t\begin{pmatrix}a(t)\\ b(t)\end{pmatrix} = -\frac{i}{2}\underbrace{\begin{pmatrix}\Delta & \Omega_R e^{-i\phi}\\ \Omega_R e^{+i\phi} & -\Delta\end{pmatrix}}_{M}\begin{pmatrix}a(t)\\ b(t)\end{pmatrix}$$

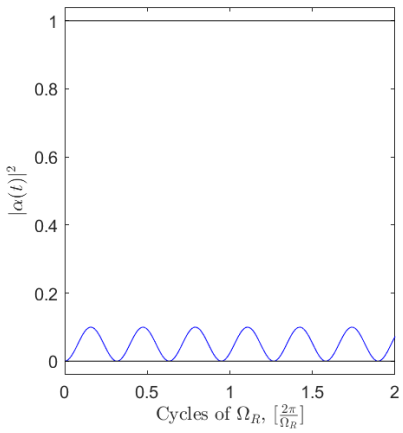
To solve this, note that $M^2 = (\Delta^2 + \Omega_R^2)I$, hence if we write a generalized frequency $\Omega = \sqrt{\Delta^2 + \Omega_R^2}$, the propagator is

$$U(t) = e^{-\frac{i}{2}Mt} = I + \left(-\frac{i}{2}t\right)M + \frac{1}{2!}\left(-\frac{i}{2}t\right)^2 \underbrace{M^2}_{\Omega^2 I} + \cdots = I\cos\left(\frac{\Omega t}{2}\right) - i\frac{M}{\Omega}\sin\left(\frac{\Omega t}{2}\right)$$

$$= \begin{pmatrix}\cos\left(\frac{\Omega t}{2}\right) - i\frac{\Delta}{\Omega}\sin\left(\frac{\Omega t}{2}\right) & -i\frac{\Omega_R}{\Omega}e^{-i\phi}\sin\left(\frac{\Omega t}{2}\right)\\ -i\frac{\Omega_R}{\Omega}e^{i\phi}\sin\left(\frac{\Omega t}{2}\right) & \cos\left(\frac{\Omega t}{2}\right) + i\frac{\Delta}{\Omega}\sin\left(\frac{\Omega t}{2}\right)\end{pmatrix}$$

Frome here, given $\begin{pmatrix}a(0)\\ b(0)\end{pmatrix} = \begin{pmatrix}0\\ 1\end{pmatrix}$, $\begin{pmatrix}a(t)\\ b(t)\end{pmatrix} = U(t)\begin{pmatrix}0\\ 1\end{pmatrix}$

$$a(t) = -i\frac{\Omega_R}{\Omega}e^{-i\phi}\sin\left(\frac{\Omega t}{2}\right) \Rightarrow |a(t)|^2 = \left(\frac{\Omega_R}{\Omega}\right)^2\sin^2\left(\frac{\Omega t}{2}\right)$$



Nice presentation of the rotation on the Bloch sphere:

$$\partial_t|\bar{\psi}\rangle = -\frac{i}{2}\begin{pmatrix}\Delta & \Omega_R e^{-i\phi}\\ \Omega_R e^{i\phi} & -\Delta\end{pmatrix}|\bar{\psi}\rangle = -\frac{i}{2}(\Omega_R\cos(\phi), \Omega_R\sin(\phi), \Delta)\cdot\vec{\sigma}|\bar{\psi}\rangle$$

Which can be written as a Hamiltonian in the rotating frame H_{rf},

$$H_{rf} = \frac{\hbar}{2}(\Omega_R\cos(\phi), \Omega_R\sin(\phi), \Delta)\cdot\vec{\sigma}$$



By defining $\cos\theta = \frac{\Delta}{\Omega}$, $\sin\theta = \frac{\Omega_R}{\Omega}$

we get $H_{rf} = \frac{\hbar\Omega}{2}\begin{pmatrix}\cos\theta & \sin\theta\, e^{-i\phi}\\ \sin\theta\, e^{i\phi} & -\cos\theta\end{pmatrix} = \frac{\hbar\Omega}{2}(\sin(\theta)\cos(\phi), \sin(\theta)\sin(\phi), \cos(\theta))\cdot\vec{\sigma}$



$$H_{rf} = \frac{\hbar\Omega}{2}(\hat{n}\cdot\vec{\sigma})$$

Thus, for a complex-valued $r = 2\Omega_R e^{i\phi}e^{-i\omega t}$, $\phi = 0, \pi$ result in pure $\pm\sigma_x$ and $\phi = \pm\frac{\pi}{2}$ result in pure $\pm\sigma_y$.

Summary:

- An effective rotation Hamiltonian is just set by choosing a convenient reference energy (removing the mean energy, $\bar{E}$ or $\bar{\omega}$)
- An effective rotation Hamiltonian is $\hbar(\vec{n}\cdot\vec{\sigma})$, where $\vec{n}$ has units of Hz (rate)
  - $H = (\vec{n}\cdot\vec{\sigma}) = \hbar\begin{pmatrix} -\frac{\nu}{2} & \frac{r^*}{\hbar} \\ \frac{r}{\hbar} & \frac{\nu}{2} \end{pmatrix}$
  - $\vec{n} = (n_x, n_y, n_z) = \left(\Re e\left[\frac{r}{\hbar}\right], \Im m\left[\frac{r}{\hbar}\right], -\frac{\nu}{2}\right)$
  - $H_0 = -\hbar\frac{\nu}{2}\sigma_z$ diagonal terms
  - $H_1 = \hbar(n_x, n_y, 0)\cdot\vec{\sigma}$ off diagonal terms
- We defined two rotating frames:
  - The <u>Heisenberg picture</u>, a frame "rotating" with the Hamiltonian.
    - Relation to Schrödinger picture is $|\tilde{\psi}\rangle = e^{\frac{i}{\hbar}H_0 t}|\psi\rangle$
    - The relevant term to calculate $e^{\frac{i}{\hbar}H_0 t}H_1 e^{-\frac{i}{\hbar}H_0 t}$
    - The state dynamics in this frame is $\partial_t|\tilde{\psi}\rangle = \frac{1}{i\hbar}e^{\frac{i}{\hbar}H_0 t}H_1 e^{-\frac{i}{\hbar}H_0 t}|\tilde{\psi}\rangle$
  - The frame that rotates with the driving field (ω), <u>the rotating frame</u>
    - Relation to Schrödinger picture is $|\bar{\psi}\rangle = e^{-i\frac{\omega t}{2}\sigma_z}|\psi\rangle$
    - The relevant term to calculate $e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z}$
    - The state dynamics in this frame is
    $$\partial_t|\bar{\psi}\rangle = \frac{1}{i\hbar}e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z}|\bar{\psi}\rangle$$
  - It is convenient to use the state in the rotating frame (with the field) however, the state is stationary only in the Heisenberg picture.
  - Here, these frames are marked $|\bar{\psi}\rangle$, $|\tilde{\psi}\rangle$, but we would not mark that explicitly in the future, but just mark $|\psi\rangle$!

- Using Rabi frequency terminology
  - Rabi frequency is the coupling term (off diagonal)
  $$\frac{r}{\hbar} = \frac{1}{2}\Omega_R e^{i\phi}$$
  - Without detuning, $\phi = 0 \rightarrow H_{rf} \propto \sigma_x$ and $\phi = \frac{\pi}{2} \rightarrow H_{rf} \propto \sigma_y$.
  -
  - The detuning is the shift of the coupling from the transition resonance, for a <u>phasor excitation</u> $r(t) = r(0)e^{-i\omega t}$
  $$\Delta = \omega - \nu$$

- The effective oscillation frequency is

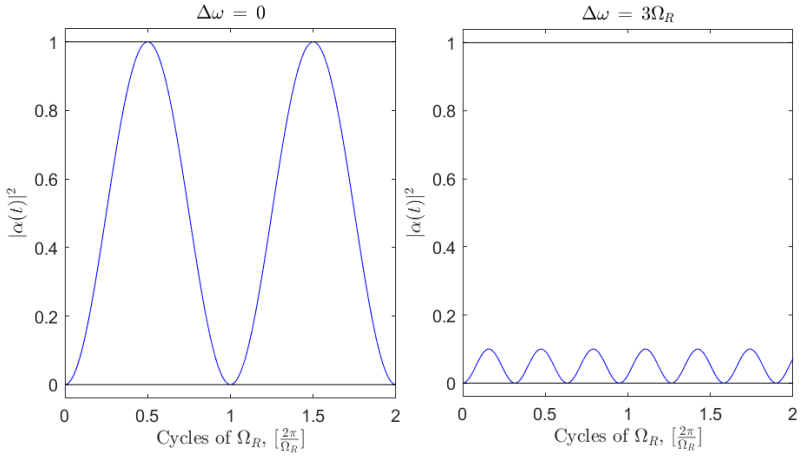
$$\Omega = \sqrt{\Delta^2 + \Omega_R^2}$$

- The probability of Rabi-oscillation amplitude is

$$\left(\frac{\Omega_R}{\Omega}\right)^2 = \frac{\Omega_R^2}{\Delta^2 + \Omega_R^2}$$

hence, detuning

  - Reduces the oscillation amplitude
  - Increases oscillation frequency ($\Omega \geq \Omega_R$).



Rotating wave approximation

If $r(t) = r(0)\cos(\omega t + \theta)$ it can be written as

$$r(t) = \frac{r(0)}{2}\left(e^{i\omega t + i\theta} + e^{-i\omega t - i\theta}\right)$$

Thus, the coupling term will be $\frac{r(0)}{2}\left(e^{i(\omega+\nu)t+i\theta} + e^{-i(\omega-\nu)t-i\theta}\right)$

If $(\omega + \nu)$ is much larger than any rate of change in our system, the contribution of its fast oscillations vanishes and we can approximate $r(t)e^{i\nu t} \approx r(0)e^{-i(\omega-\nu)t-i\theta} = r(0)e^{-i\Delta t - i\theta}$.

For example if $\omega - \nu = \Delta = 0$, and $\omega = 2\pi \cdot 10\ GHz$, then $\frac{r(0)}{2}e^{-i\theta}\left(e^{i\ \overbrace{(\omega+\nu)}^{20\ GHz\cdot 2*pi}\ t+2i\theta} + 1\right)$ and any integral over a nanosecond ($10^{-9}\ s$) the first term will nullify.

This approximation treats a real (and real-valued) field as if it is rotating in the complex plane.

→ a rotating-wave approximation.

$$r(0)\cos(\omega t + \theta) \approx \frac{r(0)}{2}e^{-i\theta}e^{-i\omega t}$$

$$r(0)\cos(\omega t + \theta)\, e^{i\nu t} \approx \frac{r(0)}{2}e^{-i\theta}e^{-i\Delta t}$$

Or in terms of cos/sin

$$r(0)\cos(\omega t) \approx \frac{r(0)}{2}e^{-i\omega t} \qquad r(0)\sin(\omega t) \approx -i\frac{r(0)}{2}e^{-i\omega t}$$

$r(0)\cos(\omega t)\,e^{i\nu t} \approx \frac{r(0)}{2}e^{-i\Delta t}$ $\qquad$ $r(0)\sin(\omega t)\,e^{i\nu t} \approx -i\frac{r(0)}{2}e^{-i\Delta t}$

Applying σ_x vs. σ_y using a real-valued harmonic field

Lets assume $H_0 = -\hbar\frac{\nu}{2}\sigma_z$ and a real-valued time dependent off-diagonal $H_1 = V_0\cos(\omega t)\sigma_x$.

We calculate $\frac{1}{i\hbar}e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z}$ and use that to write $\partial_t|\bar{\psi}\rangle$, H_{rf}.

$$\frac{1}{i\hbar}e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z} = \frac{V_0\cos(\omega t)}{i\hbar}\begin{pmatrix} e^{-i\frac{\omega t}{2}} & 0 \\ 0 & e^{i\frac{\omega t}{2}}\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix} e^{i\frac{\omega t}{2}} & 0 \\ 0 & e^{-i\frac{\omega t}{2}}\end{pmatrix}$$

$$= \frac{V_0\cos(\omega t)}{i\hbar}\begin{pmatrix}0 & e^{-i\omega t}\\ e^{i\omega t} & 0\end{pmatrix} = \frac{V_0}{i\hbar}\begin{pmatrix} 0 & \frac{1}{2}\left(e^{i\omega t}+e^{-i\omega t}\right)e^{-i\omega t}\\ \frac{1}{2}\left(e^{i\omega t}+e^{-i\omega t}\right)e^{i\omega t} & 0\end{pmatrix}.$$

Here, the rotating wave approximation is valid if we can neglect the contribution of $e^{2i\omega t}, e^{-2i\omega t}$, such that

$$\frac{1}{i\hbar}e^{-i\frac{\omega t}{2}\sigma_z}H_1 e^{i\frac{\omega t}{2}\sigma_z} \approx -i\frac{V_0}{2\hbar}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = -i\frac{V_0}{2\hbar}\sigma_x$$

Had the off-diagonal part of the Hamiltonian been sinusoidal (with σ_x) , $H_1 = V_0\sin(\omega t)\sigma_x$.

The last part would be $= \frac{V_0}{i\hbar}\begin{pmatrix} 0 & \frac{i}{2}\left(e^{i\omega t}-e^{-i\omega t}\right)e^{-i\omega t}\\ \frac{i}{2}\left(e^{i\omega t}-e^{-i\omega t}\right)e^{i\omega t} & 0\end{pmatrix} \approx -i\frac{V_0}{2\hbar}\begin{pmatrix}0 & i\\ -i & 0\end{pmatrix} =$

$-i\frac{V_0}{2\hbar}(-\sigma_y)$

We use the previous derivation $\partial_t|\bar{\psi}\rangle = \left(-i\frac{\Delta}{2}\sigma_z + \frac{1}{i\hbar}e^{i\frac{\omega t}{2}\sigma_z}H_1 e^{-i\frac{\omega t}{2}\sigma_z}\right)|\bar{\psi}\rangle$ and $H_{rf}|\bar{\psi}\rangle = i\hbar\partial_t|\bar{\psi}\rangle$ to get

for $H_1 = V_0\cos(\omega t)\boldsymbol{\sigma_x}$, $H_{rf} = \hbar\frac{\Delta}{2}\sigma_z + \hbar\frac{V_0}{2\hbar}\boldsymbol{\sigma_x} = \frac{\hbar}{2}\begin{pmatrix}\Delta & \frac{V_0}{\hbar}\\ \frac{V_0}{\hbar} & -\Delta\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}\Delta & \Omega_R\\ \Omega_R & -\Delta\end{pmatrix}$

and for $H_1 = V_0\sin(\omega t)\boldsymbol{\sigma_x}$, $H_{rf} = \hbar\frac{\Delta}{2}\sigma_z + \hbar\frac{V_0}{2\hbar}(-\boldsymbol{\sigma_y}) = \frac{\hbar}{2}\begin{pmatrix}\Delta & i\frac{V_0}{\hbar}\\ -i\frac{V_0}{\hbar} & -\Delta\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}\Delta & i\Omega_R\\ -i\Omega_R & -\Delta\end{pmatrix}$.

where we used $\Omega_R = V_0/\hbar$. The timing of the oscillation governs ϕ. If $\cos(\omega t)$ results in σ_x, then $\sin(\omega t)$ results in $-\sigma_y$.

An effective qubit from a non-two-level-system

Most systems that are used for qubits have more than two levels. So how can they be used as two level system? Why there's no effect from the other levels?

Next, we'll take the simplest multi-level system, choose two of its states as the qubit sub-space, and see why the role played by the other states.

Reminder of an infinite well potential in 1D

The potential for an electron in an infinite well

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & else \end{cases}$$

The Hamiltonian, and its eigenvalue equation for the energy is

$$H = \frac{1}{2m_e}\hat{P}^2\psi(x) + V(\hat{x})\psi(x) = -\frac{\hbar^2}{2m_e}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x), \qquad H\psi = \underbrace{\mathcal{E}}_{energy\ scaler}\psi$$

Within the well, $0 < x < L$,

$$-\frac{\hbar^2}{2m_e}\frac{d^2\psi(x)}{dx^2} = \mathcal{E}\psi(x)$$

A general solution

$$\psi = Ae^{ikx} + Be^{-ikx\rightarrow} - \frac{\hbar^2}{2m_e}\frac{d^2\psi}{dx^2} = \frac{\hbar^2k^2}{2m_e}\psi = \mathcal{E}\psi \rightarrow \mathcal{E} = \frac{\hbar^2k^2}{2m_e}$$

Boundary Conditions:

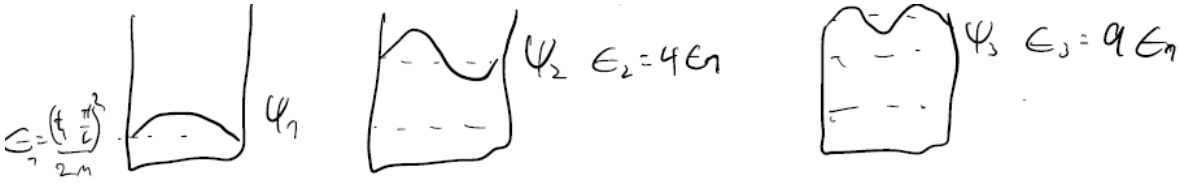
$$\psi(x=0) = 0 \rightarrow \begin{cases} A + B = 0 \\ \psi = A\left(e^{ikx} + e^{-ikx}\right) = 2A\sin(kx) \end{cases}$$

$$\psi(x=L) = 0 \rightarrow \sin(kL) = 0 \rightarrow k = \frac{n\pi}{L}$$

Normalization:

$$\psi_n = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi n}{L}x\right) \quad , \quad \mathcal{E}_n = \frac{\hbar^2\pi^2}{2m_eL^2}n^2$$

This solution uses $\psi(x=0) = \psi(x=L) = 0$ since $V_{(x<0, x>L)} \rightarrow \infty$.



The eigenstates are vectors $|\psi_n\rangle$, of which the spatial *wavefunction* is $\psi_n(x) = \langle x|\psi_n\rangle$.

Let us consider only the first two states as a two-dimensional Hilbert space, with

$$|\psi_1\rangle \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\psi_2\rangle \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Remember that the matrix representation of an operator $\hat{O}$ is $O_{mn} = \langle\psi_m|\hat{O}|\psi_n\rangle$. So the Hamiltonian $H_{mn} = \langle\psi_m|\hat{H}|\psi_n\rangle$.

The unperturbed Hamiltonian is $H = \begin{pmatrix} \mathcal{E}_1 & \\ & \mathcal{E}_2 \end{pmatrix}$, and if we define $\bar{\mathcal{E}} = \frac{(\mathcal{E}_1 + \mathcal{E}_2)}{2}$ and $\hbar\nu = \mathcal{E}_2 - \mathcal{E}_1$ as the transition frequency. So if we consider the energy with respect to $\bar{\mathcal{E}}$ we get

$$H_0 = \frac{\hbar}{2}\begin{pmatrix} -\nu & \\ & \nu \end{pmatrix} = -\frac{\hbar\nu}{2}\sigma_z$$

As above, we can write $H = H_0 + H_1$, where H_1 holds the off-diagonal terms of $\hat{H}$, $(H_1)_{12} = \langle\psi_1|H_1|\psi_2\rangle = r^*$ and $(H_1)_{21} = \langle\psi_2|H_1|\psi_1\rangle = \langle\psi_1|H_1|\psi_2\rangle^* = r$

$$H_0 = -\frac{\hbar\nu}{2}\sigma_z \quad , \quad H_1 = \begin{pmatrix} 0 & r^* \\ r & 0 \end{pmatrix}$$

We'll find r using the potential of an oscillatory field. The potential energy

$$V(x,t) = \underbrace{-eE_0 cos(\omega t)}_{Force} \cdot x$$

And we can substitute

$$r = \langle\psi_1|H|\psi_2\rangle = \langle\psi_1|H_0 + V(x,t)|\psi_2\rangle = \langle\psi_1|V(x,t)|\psi_2\rangle = \cos(\omega t)\,(-e)E_0\langle\psi_1|x|\psi_2\rangle$$

The term $\langle\psi_1|x|\psi_2\rangle$ is called the dipole matrix element, sometimes marked d. Using it explicitly,

$$r^* = \frac{2}{L}\cos(\omega t)\,(-e)E_0\int_0^L \sin\left(\frac{\pi x}{L}\right)x\sin\left(\frac{2\pi x}{L}\right)dx = \frac{2}{L}\cos(\omega t)\,(-e)E_0\left(-\frac{8}{9}\frac{L^2}{\pi^2}\right)$$

$$r(t) = \frac{16}{9\pi^2}eE_0L\cos(\omega t) \quad , V_0 = \frac{16}{9\pi^2}eE_0L$$

$$H_1 = \begin{pmatrix} 0 & V_0\cos(\omega t) \\ V_0\cos(\omega t) & 0 \end{pmatrix} = V_0\cos(\omega t)\,\sigma_x$$

So in the rotating frame, $|\bar{\psi}\rangle = e^{-i\frac{\omega t}{2}\sigma_z}|\psi\rangle$, as in the previous segment,

$$\partial_t|\bar{\psi}\rangle = -\frac{i}{2}\begin{pmatrix} \Delta & \Omega_R \\ \Omega_R & -\Delta \end{pmatrix}|\bar{\psi}\rangle \quad , \quad H_{rf} = \frac{\hbar}{2}\begin{pmatrix} \Delta & \Omega_R \\ \Omega_R & -\Delta \end{pmatrix}$$

We got our rotation (mixing σ_x, σ_z) in the subspace of $|\psi_1\rangle, |\psi_2\rangle$ as our qubit.

Parasitic coupling

The infinite well is not a two-level system. There are an infinite number of states. The oscillating field now couples more than just levels $|0\rangle \leftrightarrow |1\rangle \quad or \; |\psi_1\rangle \leftrightarrow |\psi_2\rangle$, there is also a parasitic coupling to a higher level, $|\psi_2\rangle \leftrightarrow |\psi_3\rangle$.

In this section we create a Rabi-oscillation between $|\psi_1\rangle \leftrightarrow |\psi_2\rangle$, and ask what happens to the third level. We work on resonance, $\Delta = 0$, for simplicity.

The states are represented by

$$|\psi_1\rangle \to \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |\psi_2\rangle \to \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |\psi_3\rangle \to \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

where the unperturbed Hamiltonian is $H_0 = \begin{pmatrix} -\frac{\hbar}{2}\nu & 0 & 0 \\ 0 & \frac{\hbar}{2}\nu & 0 \\ 0 & 0 & \alpha\frac{\hbar}{2}\nu \end{pmatrix}$

since the coupling is different for the different levels, we can write

$$H_1 = \begin{pmatrix} 0 & V_{12}\cos(\omega t) & 0 \\ V_{12}\cos(\omega t) & 0 & V_{23}\cos(\omega t) \\ 0 & V_{23}\cos(\omega t) & 0 \end{pmatrix}$$

Now, for the Heisenberg picture, an assistive calculation,

$$e^{\frac{i}{\hbar}H_0 t} = \begin{pmatrix} e^{-i\frac{\nu}{2}t} & 0 & 0 \\ 0 & e^{i\frac{\nu}{2}t} & 0 \\ 0 & 0 & e^{\alpha i\frac{\nu}{2}t} \end{pmatrix} = \begin{pmatrix} e^{-i\lambda} & 0 & 0 \\ 0 & e^{i\lambda} & 0 \\ 0 & 0 & e^{\alpha i\lambda} \end{pmatrix}$$

where we abbreviate $\frac{\nu}{2}t = \lambda$. We calculate $e^{\frac{i}{\hbar}H_0 t} H_1 e^{-\frac{i}{\hbar}H_0 t}$

$$\begin{pmatrix} e^{-i\lambda} & 0 & 0 \\ 0 & e^{i\lambda} & 0 \\ 0 & 0 & e^{\alpha i\lambda} \end{pmatrix} \begin{pmatrix} 0 & V_{12}\cos(\omega t) & 0 \\ V_{12}\cos(\omega t) & 0 & V_{23}\cos(\omega t) \\ 0 & V_{23}\cos(\omega t) & 0 \end{pmatrix} \begin{pmatrix} e^{i\lambda} & 0 & 0 \\ 0 & e^{-i\lambda} & 0 \\ 0 & 0 & e^{-\alpha i\lambda} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & V_{12}\cos(\omega t)\, e^{-2i\lambda} & \\ V_{12}\cos(\omega t)\, e^{2i\lambda} & 0 & V_{23}\cos(\omega t)\, e^{-(\alpha-1)i\lambda} \\ & V_{23}\cos(\omega t)\, e^{(\alpha-1)i\lambda} & 0 \end{pmatrix}$$

And in the rotating wave approximation, with no detuning, $\omega = \nu$,

$$\cos(\omega t)\, e^{2i\lambda} = \cos(\omega t)\, e^{i\nu t} \approx \frac{1}{2} e^{-i\omega t} e^{i\nu t} = \frac{1}{2} e^{-i\Delta_{12} t} \xrightarrow{no\ detuning} \frac{1}{2}$$

However, since $\left|\omega - \frac{(\alpha-1)}{2}\nu\right|$ is not much smaller than $\left|(-\omega) - \frac{(\alpha-1)}{2}\nu\right|$, the rotating-wave-approximation (RWA) is not well justified. To put some numbers for $\alpha \approx 4$ (see appendix), the RWA requires that $\left|1 - \frac{3}{2}\right| = \frac{1}{2} \ll \left|1 + \frac{3}{2}\right| = \frac{5}{2}$, which and is not met.
Hence, for the terms of V_{23} one needs to keep the terms $V_{23}\cos(\omega t)$.

The dynamics (Heisenberg picture) is

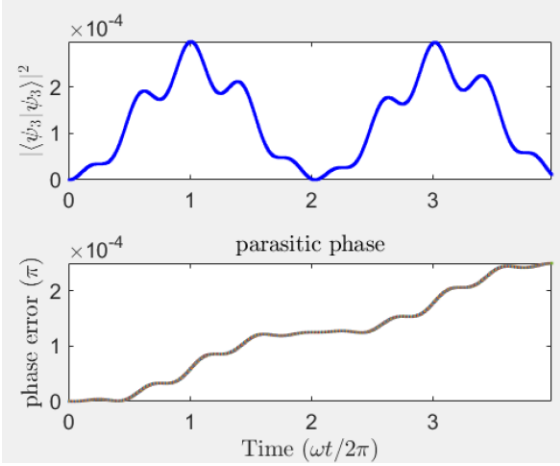
$$\partial_t |\tilde{\psi}\rangle = \frac{1}{i\hbar} e^{\frac{i}{\hbar}H_0 t} H_1 e^{-\frac{i}{\hbar}H_0 t} |\tilde{\psi}\rangle$$

$$\partial_t |\tilde{\psi}\rangle = -\frac{i}{2} \begin{pmatrix} 0 & \frac{V_{12}}{\hbar} & \\ \frac{V_{12}}{\hbar} & 0 & \frac{V_{23}}{\hbar} 2\cos(\omega t)\, e^{-\frac{3}{2}i\nu t} \\ & \frac{V_{23}}{\hbar} 2\cos(\omega t)\, e^{\frac{3}{2}i\nu t} & 0 \end{pmatrix} |\tilde{\psi}\rangle$$

The subspace of $|\psi_1\rangle, |\psi_2\rangle$ (blue block) acts like a qubit, with the intended Rabi oscillations.

The subspace of $|\psi_2\rangle, |\psi_3\rangle$ (black block, no RWA) is the parasitic coupling, which one should aspire to minimize.

So, one needs to check that no significant change can develop in the system of $|\psi_2\rangle, |\psi_3\rangle$.



Hence – the infinite square potential well is a good system to serve as an effective qubit.

Appendix

calculation of the eigen energies and coupling terms for 3 levels of a particle in a box

We now mark the transition energy between levels m, n as ν_{mn}, where up to now we used only ν_{12}.

Let's examine their role by looking at the transition between $|\psi_2\rangle \leftrightarrow |\psi_3\rangle$, as well when we excite using the $\omega = \nu_{12}$.

(*) BTW – is there a coupling $|\psi_1\rangle \leftrightarrow |\psi_3\rangle$? Why?

Note that $\mathcal{E}_n = \frac{\hbar^2\pi^2}{2m_e L^2} n^2 = \mathcal{E}_1 n^2$, so

$$\nu_{12} = \mathcal{E}_1(4-1) = 3\mathcal{E}_1 \quad , \qquad \nu_{23} = 5\mathcal{E}_1$$

How far are each of these energies from our soon-to-be new zero, the energy $\bar{\mathcal{E}} = \frac{\mathcal{E}_1+\mathcal{E}_2}{2} = \mathcal{E}_1$?
We'll be using our marking of $\mathcal{E}_2 - \mathcal{E}_1 = 3\mathcal{E}_1 = \hbar\nu$.

1st level $\rightarrow \mathcal{E}_1 - \bar{\mathcal{E}} = -\hbar\nu/2$, 2nd level $\mathcal{E}_2 - \bar{\mathcal{E}} = \hbar\nu/2$,
and the 3rd level, $\mathcal{E}_3 - \bar{\mathcal{E}} = \mathcal{E}_3 - \mathcal{E}_2 + \hbar\nu\frac{1}{2} = 5\mathcal{E}_1 + \hbar\nu\frac{1}{2} = \hbar\nu\left(\frac{5}{3}+\frac{1}{2}\right) = \frac{13}{6}\hbar\nu = \alpha\frac{1}{2}\hbar\nu$.

$\alpha = \frac{13}{3} \approx 4$.

In this three-dimensional state basis is

$$|\psi_1\rangle \rightarrow \begin{pmatrix}1\\0\\0\end{pmatrix}, |\psi_2\rangle \rightarrow \begin{pmatrix}0\\1\\0\end{pmatrix}, |\psi_3\rangle \rightarrow \begin{pmatrix}0\\0\\1\end{pmatrix}$$

Here, Pauli matrices are no longer the complete set of Unitary Hermitian matrices. The rotation generators are L_x, L_y, L_z.

For the coupling $|\psi_1\rangle \leftrightarrow |\psi_2\rangle$, we found $\langle\psi_1|V(x,t)|\psi_2\rangle = \cos(\omega t)\,(-e)E_0\langle\psi_1|x|\psi_2\rangle = \frac{16}{9\pi^2} eE_0 L\cos(\omega t) = V_{12}\cos(\omega t)$, with $V_{12} = \frac{16}{9\pi^2} eE_0 L$.

Now the field can also couple $|\psi_2\rangle \leftrightarrow |\psi_3\rangle$, with a term V_{23}.
One can calculate

$$\langle\psi_2|V(x,t)|\psi_3\rangle = \cos(\omega t)\,(-e)E_0\langle\psi_2|x|\psi_3\rangle = \frac{48}{25\pi^2} eE_0 L\cos(\omega t) = V_{23}\cos(\omega t)$$

Hence , through the dipole matrix element $V_{23} = \frac{48}{25\pi^2} eE_0 L$.

Rotation by a Hamiltonian

```
hbar=1;

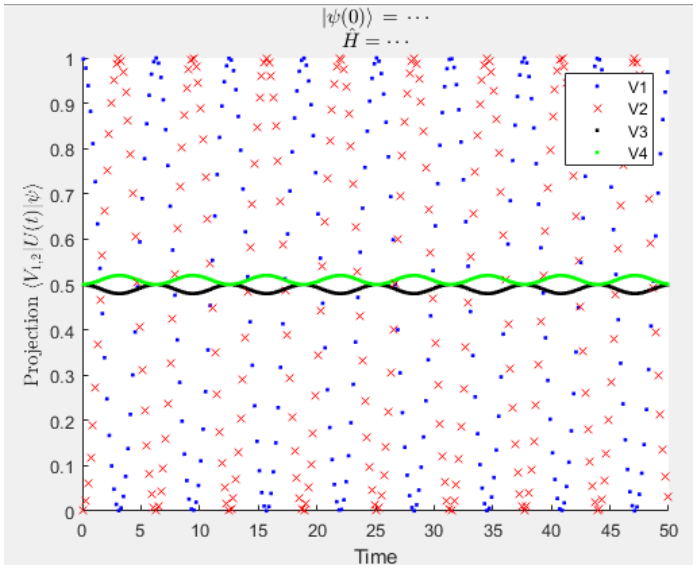
V1=[1,1]';
V2=[1,-1]';
V3=[1,0]';
V4=[0,1]';
V1=V1./norm(V1);
V2=V2./norm(V2);
V3=V3./norm(V3);
V4=V4./norm(V4);
Psi=V1;
```

```
Psi=Psi./norm(Psi);


E1=-0.5;
E2=0.5;
e=0.01;

H=[E1, e ;
    conj(e), E2];


figure(1); clf
for t=0.1:0.2:50;
t_final=t;
U=expm(-1i./hbar.*H*t_final);
hold on;
plot(t_final, abs(V1'*U*Psi).^2,'b.',t_final,
abs(V2'*U*Psi).^2,'rx')
plot(t_final, abs(V3'*U*Psi).^2,'k.',t_final,
abs(V4'*U*Psi).^2,'g.')
xlim([0,50]);
xlabel('Time')
drawnow;
pause(0.01);
end
hold off
ylim([0,1]);
ylabel('Projection $\langle V_{1,2}|U(t) |\psi \rangle$','Interpreter','latex')
title({'$|\psi(0)\rangle \, = \, \cdots$','$\hat{H}=\cdots$'},'Interpreter','latex')
legend('V1','V2','V3','V4')
%%H\
```



Rabi-Oscillation calculation

Matlab code:

```
% This plots several curves for Rabi oscillations
detuning_vec=[0,1,4];

for kk=1:length(detuning_vec)
    subplot(3,1,kk)

Omega_R=1;
delta_omega=detuning_vec(kk);
Omega=sqrt(Omega_R^2+delta_omega.^2);
sin_theta=Omega_R./Omega;
cos_theta=delta_omega./Omega;

t=(0:0.01:2).*2*pi./Omega_R;

prob_alpha=sin_theta.^2.*(sin(Omega.*t/2)).^2;
prob_beta=sin_theta.^2.*(cos(Omega.*t/2)).^2;
t_axis=Omega_R.*t/(2*pi);
plot(t_axis,prob_alpha,'b',t_axis,t_axis.*0,'k',t_axis,t_axis.*0+1,'k')
hold on;
plot(t_axis,prob_beta,'r',t_axis,t_axis.*0,'k',t_axis,t_axis.*0+1,'k')
hold off;
ylim([-0.04,1.04]);
title(['$\Delta\omega\, =\, ',num2str(round(delta_omega,2)),' \Omega_R$'],'Interpreter','latex')
if kk==3;
xlabel('Cycles of $\Omega_R$, $[\frac{2 \pi}{\Omega_R}]$','interpreter','latex')
end
if kk==2;
    ylabel('$|\alpha(t)|^2$ (blue) and $|\beta(t)|^2$ (red)','Interpreter','latex')
end

end
```

Effective qubit within a 3-level system

Matlab code: (more visualized code in the comment)

```
%% Evolution of the parasitic (3rd) state for an effective qubit within a 3-level system
hbar=1;


V1=[1,1,0]';
V2=[-1,1,0]';
V3=[0,0,1]';
V1=V1./norm(V1);
V2=V2./norm(V2);
V3=V3./norm(V3);



V12=0.05;
V23=0.05;
w=5;
nu=5;
alpha = 13/3;


Psi3_t=V1/norm(V1);



V11=V1(1:2);
Psi2_t=V11/norm(V11);



figure(1); clf
dt=0.005;
t_max=5;


for t=0.01:dt:t_max;



subplot(2,1,1)
plot(w*t/(2*pi), abs(V3'*Psi3_t).^2,'b.')
hold on;
ylabel('$|\langle \psi_3 | \psi_3 \rangle|^2 $','Interpreter','latex')
xlim([0,w*t_max/(2*pi)]);
set(gca,'Fontsize',14)


subplot(2,1,2)
plot(w*t/(2*pi), angle(Psi2_t'*Psi3_t(1:2))/pi,'r.')
ylabel('phase error ($\pi$)','Interpreter','latex')
hold on;
set(gca,'Fontsize',14)


xlim([0,w*t_max/(2*pi)]);
xlabel('Time ($\omega t / 2\pi)$','Interpreter','latex')
drawnow;
pause(0.01);

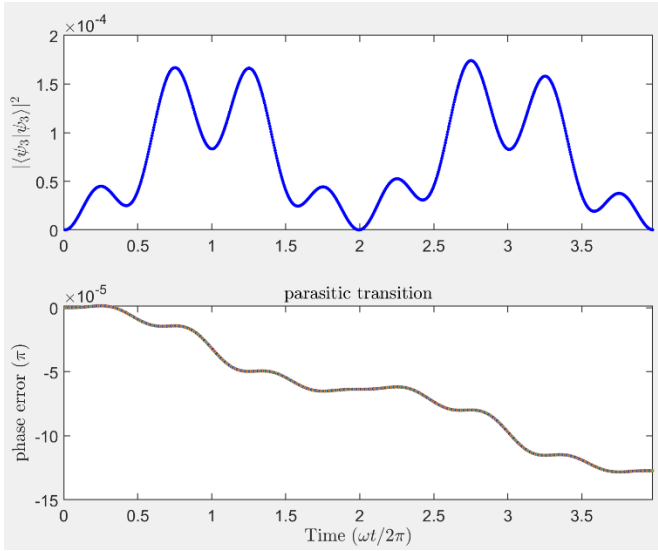

H23=V23.*cos(w.*t).*exp(-1i*  (alpha-1) /2   *nu.*t);


H=  [ 0    ,   V12    ,   0 ;...
      V12  ,   0      , H23; ...
       0   ,  conj(H23), 0  ];


Psi3_t  =  Psi3_t  -  dt .* 1i .* H * Psi3_t;
Psi2_t  =  Psi2_t  -  dt .* 1i .* H(1:2,1:2) * Psi2_t;


end
hold off


title({'parasitic phase'},'Interpreter','latex')
%%H\
```



Adiabatic single qubit gate for a δ-well pair

Matlab code:

```
%% adiabatic double-waveguide gate
draw_final_only=true;

t=(-10:0.1:10);
% parameters for X gate:
% k=.9999; dn=1.58; df=8/k;
% parameters for 5050 gate:
% k=.9999; dn=2.3136; df=8/k;
k=.9999; dn=0.5; df=8/k;


d=df-(df-dn).*0.5*(1+tanh(t+5)).*0.5.*(1-tanh(t-5));
subplot(2,1,1);
r=exp(-k.*d);
p1=plot(t, [d;-d],'b', t,10*r,'r',t,t*0,'k' );
legend('Well separation, d', 'coupling r (or \Omega_R )')

% syms rs
% H=[0 rs; conj(rs) 0];

A=t*NaN; % these are the amplitudes alpha and beta
B=A;
A0=0;
B0=1;
dt=t(2)-t(1);
psi0=[0;1];

subplot(2,1,2);

A(1)=A0;
B(1)=B0;

p2=plot(t,t*0,'k',t,t*0+1,'k',t(1), abs(A0).^2,'ro', t(1), abs(B0).^2,'bo');
xlabel('$time (arb.u)$','interpreter','latex');
ylabel('$|\alpha(t|^2$ and $|\beta(t|^2$','interpreter','latex');
hold on;
tic;
for kt=1:length(t)-1
Hkt=[0 r(kt); conj(r(kt)) 0];
tmp_v_before    =   [A(kt); B(kt)];
tmp_v_after     =   expm(-1i*Hkt.*dt)*tmp_v_before;
A(kt+1)=tmp_v_after(1);
B(kt+1)=tmp_v_after(2);


switch draw_final_only
    case false
    plot(t(kt+1),abs(A(kt+1)).^2,'r.',t(kt+1),abs(B(kt+1)).^2,'b.');
    drawnow;
    pause(0.005);

    case true
end



end
toc;

plot(t,abs(A).^2,'r',t,abs(B).^2,'b');
hold off;
legend('$|\alpha(t|^2$','$|\beta(t|^2$','interpreter','latex');
disp(abs(B(end)).^2)
```